

# Minerva: GPU-Accelerated HPC Infrastructure for Biomedical AI and Scientific Computing

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**Abstract**— The rapid growth of AI-driven applications, including LLMs, genomics, medical imaging, and data-intensive biomedical research, has significantly increased the demand for HPC infrastructure. This paper presents an overview of the Minerva HPC system at the Icahn School of Medicine at Mount Sinai Health System, a large-scale HIPAA-compliant supercomputing environment designed to support biomedical and AI-driven workloads. Minerva integrates heterogeneous CPU and GPU resources, including NVIDIA V100, A100, H100, B200, and L40S GPUs, together with NVIDIA NDR InfiniBand networking, NVLink interconnects, and GPFS parallel storage. We conducted experimental evaluations using the NVIDIA HPL benchmark across multiple GPU architectures. The NVIDIA H100NVL GPU achieved 165.40 TFLOPS using four GPUs, while the NVIDIA B200 GPU achieved the highest throughput of 262.50 TFLOPS using eight GPUs. The paper also discusses LLM-based ticket mining, GPU resource optimization, and AI-focused researcher training designed to improve operational efficiency, user support, and overall HPC resource utilization.

**Keywords**— *High-performance Computing (HPC), Biomedical Research, GPU Computing, HPL Benchmark, Large Language Model (LLM).*

## I. INTRODUCTION

The increasing adoption of artificial intelligence (AI), machine learning (ML), genomics, medical imaging, molecular dynamics, and large language model (LLM) applications has transformed modern biomedical research and healthcare analytics. These applications require extremely large computational resources, scalable storage systems, fast interconnect networks, and GPU-accelerated computing platforms capable of processing massive volumes of scientific and clinical data.

To address these growing computational demands, the Icahn School of Medicine at Mount Sinai Health System developed the Minerva HPC system, a large-scale HIPAA-compliant computational infrastructure designed to support biomedical and AI-driven research workloads [1]. Minerva integrates heterogeneous CPU and GPU resources, including NVIDIA V100, A100, H100, L40S, and B200 accelerators, together with NVIDIA NDR InfiniBand networking, NVLink interconnects, GPFS parallel storage, and large-memory compute nodes. Since its initial deployment in 2012, Minerva has undergone multiple upgrades and currently delivers more than 20 petaflops of computational performance. The system has supported over 2,100 peer-reviewed publications and was ranked 175 in the TOP500 list in November 2025.

This paper presents an overview of the Minerva architecture and evaluates multiple NVIDIA GPU platforms using the NVIDIA High Performance Linpack (HPL) benchmark. The H100NVL achieved 165.40 TFLOPS using four GPUs, while the B200 system achieved the highest throughput of 262.50 TFLOPS using eight GPUs. We also describe operational strategies, including LLM-based ticket mining, GPU resource optimization, and AI-focused user training.

## II. MINERVA HPC SYSTEM OVERVIEW

The Minerva high-performance computing (HPC) system at the Icahn School of Medicine at Mount Sinai Health System is a large-scale computational infrastructure designed to support biomedical research [1]. Minerva is Health Insurance Portability and Accountability Act (HIPAA) compliant, enabling secure storage and processing of Protected Health Information (PHI). Currently, Minerva delivers more than 20 petaflops of compute performance. Minerva is organized into several major computational partitions, including Chimera, AIMS, and CATS each designed to support different classes of scientific and AI workloads. The system integrates NVIDIA NDR InfiniBand networking, NVLink GPU interconnects, local NVMe SSD storage, and direct water-cooling technologies to support large-scale distributed computing and GPU-accelerated applications. Fig. 1 presents the architecture of the Minerva HPC system.

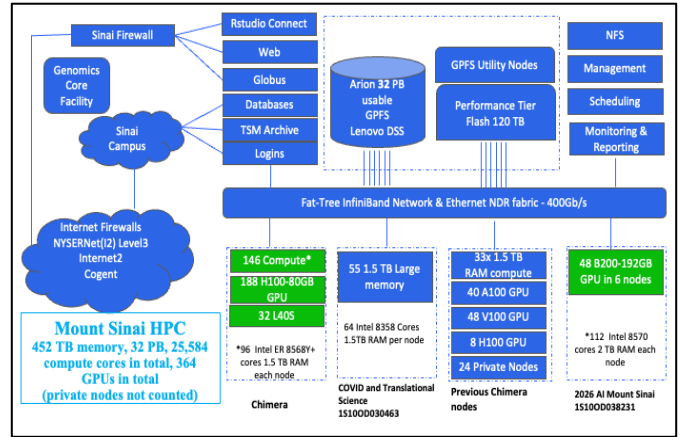


Fig. 1. In this figure, we present high-level architecture of the Minerva HPC system [2].

### A. Chimera Partition

The Chimera partition is the primary computational partition of the Minerva supercomputer and provides large-scale CPU and GPU resources for biomedical research, artificial intelligence (AI), machine learning (ML), genomics, imaging, and data-intensive scientific applications. The partition was significantly upgraded in November 2024 with the introduction of Intel Emerald Rapids processors and NVIDIA NDR InfiniBand networking. Chimera includes four login nodes based on Intel Emerald Rapids 8568Y+ processors operating at 2.3 GHz, each providing 96 cores and 512 GB of memory. The main compute infrastructure consists of 146 compute nodes equipped with Intel Emerald Rapids 8568Y+ processors, each containing 96 cores and 1.5 TB of memory, resulting in a total of 14,016 CPU cores. In addition, the partition includes 33 high-memory nodes based on Intel 8268 processors with 1.5 TB memory per node to support memory-intensive scientific workloads. The partition utilizes a 400 Gb/s NVIDIA NDR InfiniBand fat-tree network architecture together with high-speed NVMe local storage and direct water-cooling technologies to support scalable distributed computing and high-performance data processing.

The GPU infrastructure of the Chimera partition includes a heterogeneous collection of NVIDIA accelerators designed to support a broad range of AI and scientific computing applications. The GPU infrastructure is primarily composed of NVIDIA H100 GPUs, including 188 H100-80GB SXM5 GPUs distributed across 47 compute nodes and an additional 8 H100-80GB GPUs across 2 nodes. Each node contains four H100 GPUs. The system also includes NVIDIA A100 GPUs consisting of 32 A100-40GB GPUs across 8 nodes and 8 A100-80GB GPUs across 2 nodes. In addition, the partition contains 48 NVIDIA V100-16GB GPUs distributed across 12 nodes to support traditional GPU-accelerated scientific computing applications. For AI inference and visualization workloads, the system includes 32 NVIDIA L40S-48GB GPUs across 4 nodes. The NVIDIA V100 GPUs are used for traditional GPU-accelerated biomedical workloads and scientific visualization applications such as VMD, while the NVIDIA L40S GPUs support AI inference, LLM serving, generative AI, and others workloads.

### B. AIMS Partition

The AIMS partition represents the latest AI-focused expansion of Minerva and was launched in February 2026 through a \$2 million NIH S10 award. The AIMS partition is specifically designed to support large-scale AI training, large language models (LLMs), biomedical foundation models, and GPU-accelerated scientific computing. The AIMS partition consists of six Lenovo SR780a V3 DGX systems containing a total of 48 NVIDIA B200 GPUs. Each node includes eight NVLink B200 GPUs with 192 GB of HBM3e memory per GPU, resulting in approximately 9 TB of total GPU memory across the partition. Each node is additionally equipped with 112 Intel Xeon Platinum 8570 processors operating at 2.1 GHz, 2 TB of system memory, and 25 TB of local NVMe SSD storage. The B200 GPUs support FP4 (4-bit floating point) computation, enabling near-exaflop-scale AI inference performance for large-scale AI workloads.

### C. CATS Partition

The CATS partition was funded through a separate \$2 million S10 NIH award and is designed to support memory-intensive computational workloads. The partition consists of 55 nodes based on Intel Ice Lake 8358 processors operating at 2.6 GHz. Each node contains 64 cores and 1.5 TB of memory, providing a total of 3,520 CPU cores and 82.5 TB of aggregate system memory. The large-memory configuration of the CATS partition enables efficient execution of large-scale genomics workflows, biomedical simulations, and data-intensive scientific applications.

### D. GPU Configuration

GPU resources are managed using IBM Spectrum Load Sharing Facility (LSF) with queue-based scheduling, fair-share policies, and resource throttling designed to balance interactive usage and large-scale computational workloads. GPU access is provided through dedicated *gpu* and *gpuexpress* queues, with cluster-wide limits enforced on physical GPU allocation (*ngpus\_physical*). During daytime hours, GPU usage is capped at 60 GPUs to maintain responsiveness for interactive and short-running workloads, while the limit is increased to 70 GPUs during overnight hours to maximize throughput for long-running training and batch jobs. The scheduling configuration also supports backfilling, exclusive execution policies for selected queues, and per-host job limits to prevent oversubscription. To improve overall hardware utilization, CPU-only workloads are permitted to use available CPU resources on selected GPU nodes through shared LSF host groups (*express\_shared*), while GPU allocation itself remains independently controlled through GPU-specific resource policies. In addition, GPUDirect RDMA is enabled on supported GPU nodes to reduce CPU overhead and improve inter-node GPU communication performance for distributed AI and HPC workloads.

### E. File System Storage

Minerva uses IBM General Parallel File System (GPFS) to support large-scale biomedical workflows requiring high metadata performance, tiered storage, and scalable parallel I/O. GPFS is particularly advantageous for genomics and AI applications involving a very large number of files and high-throughput data access patterns. Metadata services are stored on flash storage to improve performance for file-intensive workloads. The primary parallel file system on Minerva is the Arion storage system, accessible through */sc/arion*. The storage infrastructure combines multiple Lenovo DSS storage systems and Lenovo G201 flash storage systems, providing a total raw capacity of 46 petabytes and approximately 32 petabytes of usable storage. The Arion storage system has been incrementally expanded since 2019 to support the increasing storage requirements of biomedical AI, imaging, and genomics research workflows.

### F. Software Environment

The Minerva is designed to support a wide range of scientific and computational research activities by providing researchers with access to a comprehensive collection of software tools and libraries. With a strong emphasis on artificial intelligence (AI)

and machine learning (ML), Minerva supports leading deep learning frameworks such as PyTorch, TensorFlow, and MXNet, enabling the development, training, and deployment of complex neural network models. Minerva also supports Ollama for the efficient deployment and serving of large language models (LLMs), allowing researchers to run and manage generative AI models locally on the HPC infrastructure with GPU acceleration. In addition, advanced AI platforms such as NVFlare enable federated learning across decentralized data sources while maintaining data privacy and security. In bioinformatics, Minerva offers essential tools for genomic and structural biology analyses, with AlphaFold and CryoSPARC pivotal for protein structure prediction and cryo-electron microscopy data processing, respectively. Minerva supports Amber, GROMACS, and NAMD, which are critical for understanding the behavior of biological molecules at an atomic level. Minerva also supports a variety of programming languages and utilities, including Python, R, Julia, as well as visualization tools like Matplotlib and VMD, enhancing overall computational capabilities and facilitating better interpretation and communication of results. Development and computational efficiency are enhanced by compilers like GCC and Intel Parallel Studio, containerization technologies such as Singularity, and data analysis tools like Anaconda, and RAPIDS. We provide a summary of the software available on Minerva in Table I.

TABLE I. SUMMARY OF THE POPULAR SOFTWARE AVAILABLE ON MINERVA

| Category                                     | Modules                                                                                                                                                                                                                                         |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Artificial Intelligence and Machine Learning | DeepLabCut, DeepVariant, GPT, Jax, Ollama, MXNet, MONAI, NVFlare, PyTorch, RAPIDS, Scikit-learn, TensorFlow, Torch, NVIDIA NCCL, cuDNN                                                                                                          |
| Bioinformatics                               | admixtools, afni, alphafold, alphamissense, BAM2FASTX, Bismark, BWA, bowtie, CADD, CellRanger, CNVnator, CryoSPARC, FreeSurfer, GATK, GenomeAlignmentTools, HiC-Pro, HLA, Metaphlan, MiXCR, MMSplice, PLINK, RELION, Samtools, SeqTK, STAR, VEP |
| Compilers                                    | GCC, Intel Parallel Studio, LLVM                                                                                                                                                                                                                |
| Containerization                             | Apptainer, Singularity                                                                                                                                                                                                                          |
| Data Analysis                                | Miniforge, Jupyter, R, SAS                                                                                                                                                                                                                      |
| Database Tools                               | MySQL, PostgreSQL, SQLite                                                                                                                                                                                                                       |
| Development Tools                            | Ant, Apache Maven, Gradle, Make, SCons                                                                                                                                                                                                          |
| Libraries                                    | Boost, Eigen, GSL, HDF5, LAPACK, OpenBLAS                                                                                                                                                                                                       |
| Mathematical                                 | MATLAB, R, Rmath, Mathematica                                                                                                                                                                                                                   |
| Parallel Computing                           | OpenMPI, CUDA, SLURM                                                                                                                                                                                                                            |
| Molecular Dynamics                           | Amber, CHARMM, GROMACS, NAMD, Plumed                                                                                                                                                                                                            |
| Programming Languages                        | C, C++, Fortran, Go, Java, Julia, Python, R, Rust                                                                                                                                                                                               |
| Utilities                                    | Curl, Git, Htop, Nano, Vim, Wget                                                                                                                                                                                                                |
| Version Control                              | Git                                                                                                                                                                                                                                             |
| Visualization                                | Paraview, VMD                                                                                                                                                                                                                                   |

### III. EXPERIMENTAL EVALUATIONS

This section presents the experimental evaluation of the Minerva HPC system using the NVIDIA High Performance Linpack (HPL) and NAMD3 benchmarks across multiple

NVIDIA GPU architectures. The HPL benchmark was used to evaluate and compare floating-point computing performance under compute-intensive workloads, while the NAMD3 benchmark assessed multi-GPU scaling performance for molecular dynamics simulations. Together, these benchmarks provide insight into both peak computational capability and application-level performance on modern GPU-accelerated HPC systems.

#### A. High Performance Linpack (HPL) Benchmark

The HPL benchmark is one of the most widely used benchmarks for evaluating the floating-point computational performance of high-performance computing (HPC) systems and is the primary benchmark used for the TOP500 supercomputer rankings. HPL measures system performance by solving a large dense system of linear equations using double-precision floating-point arithmetic and distributed-memory parallel processing. NVIDIA’s HPL implementation is optimized for modern GPU-accelerated architectures and supports advanced features such as GPU acceleration, MPI-based distributed execution, out-of-core memory support, and FP64 emulation for newer GPU architectures. The benchmark evaluates the efficiency of processors, GPUs, memory subsystems, and interconnect networks under highly compute-intensive workloads, making it an important tool for analyzing and comparing the peak performance of HPC clusters and AI supercomputing systems [3], [4].

#### B. Experimental Setup

We evaluated the performance of the NVIDIA High Performance Linpack (HPL) benchmark on five different GPU platforms: NVIDIA A100-40GB, A100-80GB, H100NVL, H10080G, and B200 GPUs. The V100 and L40S GPUs were not included in this study because the experiments focused on newer GPU architectures with higher FP64 computing performance designed for modern HPC workloads. The experiments were conducted using the NVIDIA HPL benchmark container. For the A100-40GB, A100-80GB, H100NVL, and H10080G GPUs, we used NVIDIA HPC-Benchmarks version 25.04. For the B200 GPU experiments, we used NVIDIA HPC-Benchmarks version 25.09 [5]. HPL is a distributed-memory benchmark designed to solve a dense system of linear equations using LU decomposition with partial pivoting. The benchmark is widely used for evaluating the floating-point computing performance of high performance computing systems. The benchmark performance is strongly influenced by parameters such as matrix size (Ns), block size (NB), process grid dimensions (P × Q), and memory utilization. The problem size parameter (Ns) was determined based on the available GPU memory using the equation shown below:

$$N_s = \xi \times 11600 \times \sqrt{\frac{M}{1GB}} \times \sqrt{C}$$

where  $N_s$  represents the HPL matrix size,  $\xi$  is the memory utilization factor,  $M$  is the GPU memory capacity, and  $C$  is the number of GPUs. In our experiments, we used approximately

80% of the total GPU memory capacity for the calculations. Table II summarizes the GPU models, and major HPL configuration parameters used in this study.

TABLE II. EXPERIMENTAL CONFIGURATION USED FOR HPL BENCHMARKING.

| GPU Type  | GPU Memory | Ns     | NB   | P × Q |
|-----------|------------|--------|------|-------|
| A100-40GB | 40 GB      | 117760 | 1024 | 2 × 2 |
| A100-80GB | 80 GB      | 165888 | 1024 | 2 × 2 |
| H100NVL   | 80 GB      | 165888 | 1024 | 2 × 2 |
| H10080G   | 80 GB      | 165888 | 1024 | 2 × 2 |
| B200      | 192 GB     | 352256 | 2048 | 4 × 2 |

For the A100-40GB GPU, the HPL.dat configuration used a matrix size (Ns) of 117760, block size (NB) of 1024, and a 2 × 2 process grid configuration. The remaining parameter values were kept at the default settings provided in the HPL.dat file included with the NVIDIA HPL benchmark container. For the A100-80GB, H100NVL, and H10080G GPUs, the same HPL.dat configuration was used except that the matrix size parameter (Ns) was increased to 165888 to utilize the larger GPU memory capacity. For the B200 GPU, a larger benchmark configuration was used to better utilize the increased memory capacity of the GPU. The HPL.dat configuration used Ns = 352256, NB = 2048, and a 4 × 2 process grid configuration. All benchmark runs were executed using the NVIDIA-provided containers to ensure software consistency across different GPU architectures.

### C. Results

Table III presents the HPL benchmark performance results obtained on the evaluated GPU platforms. The benchmark performance is reported in terms of total throughput (TFLOPS) and per-GPU throughput. The execution time required to complete the benchmark is also reported for each configuration.

The results demonstrate substantial performance differences across GPU architectures. Among the four-GPU configurations, the H100NVL GPU achieved the highest overall HPL performance of 165.40 TFLOPS with a per-GPU performance of 41.34 TFLOPS. The H100NVL system also completed the benchmark in the shortest execution time of 18.41 seconds for the same matrix size configuration used by the A100-80GB and H10080G GPUs. The A100-40GB GPU configuration achieved the lowest performance among all tested platforms, reaching 37.53 TFLOPS total performance and 9.382 TFLOPS per GPU. Increasing the GPU memory capacity from A100-40GB to A100-80GB allowed the use of a larger matrix size, resulting in improved benchmark throughput of 68.95 TFLOPS. The H10080G GPU significantly outperformed both A100 GPU configurations, achieving 113.40 TFLOPS total throughput and 28.36 TFLOPS per GPU. Compared with the A100-80GB GPU, the H10080G achieved approximately 64.5% higher total HPL performance. The B200 GPU configuration achieved the highest total benchmark throughput of 262.50 TFLOPS using eight

GPUs and a larger matrix size of 352256. The benchmark configuration for the B200 GPUs used a larger block size (NB = 2048) and a 4 × 2 process grid configuration to better utilize the larger GPU memory capacity and computational resources. The B200 system achieved 32.82 TFLOPS per GPU while processing the largest benchmark problem size among all evaluated systems.

TABLE III. HPL BENCHMARK PERFORMANCE RESULTS ACROSS DIFFERENT GPU PLATFORMS.

| GPU Type  | Ns     | NB   | GPUs Used | Time (sec.) | Total TFLOPS | Per GPU TFLOPS |
|-----------|--------|------|-----------|-------------|--------------|----------------|
| A100-40GB | 117760 | 1024 | 4         | 29.01       | 37.53        | 9.382          |
| A100-80GB | 165888 | 1024 | 4         | 44.14       | 68.95        | 17.24          |
| H100NVL   | 165888 | 1024 | 4         | 18.41       | 165.40       | 41.34          |
| H10080G   | 165888 | 1024 | 4         | 26.83       | 113.40       | 28.36          |
| B200      | 352256 | 2048 | 8         | 110.99      | 262.50       | 32.82          |

Compared with NVIDIA’s published performance results, the measured HPL benchmark results closely match the expected hardware performance. NVIDIA reports approximately 9.7 TFLOPS FP64 for the A100 GPU, 67 TFLOPS FP64 Tensor Core for the H100 series, and about 40 TFLOPS FP64 for the B200 GPU. In our experiments, the A100-40GB achieved 9.382 TFLOPS per GPU, the H10080G achieved 28.36 TFLOPS, the H100NVL achieved 41.34 TFLOPS, and the B200 achieved 32.82 TFLOPS per GPU [6], [7], [8].

Overall, the experimental results demonstrate the significant performance improvements provided by newer GPU architectures for dense linear algebra workloads. The H100NVL and B200 GPUs showed the highest HPL benchmark performance, highlighting their suitability for large-scale high performance computing applications.

### D. NAMD3 (Nanoscale Molecular Dynamics3) Benchmark

We also evaluated the scaling performance of NAMD3, a molecular dynamics simulation workload, on a single node equipped with NVLink-connected NVIDIA B200 GPUs (Fig. 2) [9]. The benchmark was conducted using one CPU core per GPU while varying the number of GPUs from 1 to 8. Performance increased from 15.6 ns/day on a single GPU to 72.8 ns/day on eight GPUs, representing a 4.67X speedup over the single-GPU configuration. Scaling was nearly linear up to four GPUs and remained strong through eight GPUs, demonstrating efficient utilization of the NVLink-connected accelerators. These results indicate that the high-bandwidth, low-latency NVLink interconnect substantially reduces inter-GPU communication overhead and plays a key role in enabling

efficient multi-GPU performance for the NAMD3 execution model.

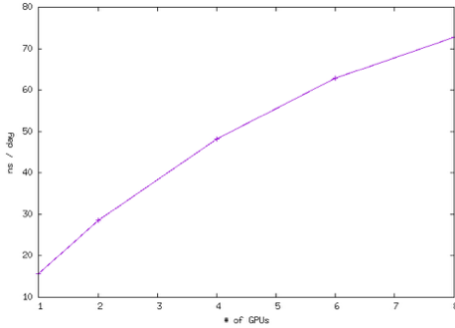


Fig. 2. Scaling performance of NAMD3 on a single node with NVLink-connected NVIDIA B200 GPUs, showing throughput increasing from 15.6 ns/day on 1 GPU to 72.8 ns/day on 8 GPUs.

#### IV. USER SUPPORT AND RESOURCE OPTIMIZATION

In this section, we describe the AI-driven operational analysis, researcher training activities, and GPU resource optimization strategies developed for the Minerva HPC resource.

##### A. LLM-Based Ticket Data Mining

The Minerva team developed an AI-powered ticket mining framework using the Llama 3.3 large language model (LLM) running through the Ollama platform on GPU-enabled Minerva infrastructure. The ticket mining workflow used a structured prompt that instructed the Llama 3.3 model to identify the primary user question, extract the key resolution provided by the HPC support team, and assign the tickets to predefined operational categories. To evaluate classification quality, a randomly selected subset of tickets was manually reviewed and compared with the LLM-generated output. The review showed high agreement between manually assigned and LLM-generated categories, demonstrating that the framework was sufficiently accurate for operational reporting and FAQ generation. The framework analyzed 5,641 unique support tickets collected between April 2024 and April 2025 and automatically categorized tickets into operational groups such as Minerva login, account management, file permission changes, package installation, package usage, GPU and AI support, LSF scheduling, containers, Jupyter, RStudio, OnDemand, storage management, and policy-related inquiries. The analysis showed that Minerva login issues (686 tickets), account-related requests (460 tickets), file and directory permission changes (452 tickets), package usage questions (377 tickets), system administration issues (378 tickets), and GPU/AI-related support requests (166 tickets) represented some of the most common operational categories. The framework also extracted the primary user question together with the corresponding solution provided by the HPC support team and identified recurring operational patterns. Based on the aggregated LLM analysis, the system automatically identified the top 30 most common and impactful user questions, which were subsequently used to update and expand the Minerva FAQ documentation. The AI-driven workflow significantly reduced

the manual effort required for ticket analysis and knowledge extraction while improving documentation quality, operational insight, and overall user support efficiency within the Minerva HPC environment. The updated FAQ page is available here [10].

##### B. Training Sessions

To help researchers effectively utilize AI and GPU resources, the Minerva team offers a series of AI-focused training sessions each year during the Spring and Fall semesters. These sessions cover topics such as: (a) Introduction to GPU/AI resources on Minerva, (b) Accelerating Biomedical Data Science with GPUs: Practical Approaches and Tools, (c) Leveraging Large Language Models in Biomedical Research, and (d) Using Containers on Minerva and Accelerating Genome Analysis with Parabricks [11]. Recordings of the training sessions are made publicly available through the Minerva website to support continuous learning and onboarding of new users. During the Spring 2026 semester, a total of 111 participants attended these sessions.

##### C. GPU Resource Optimization

The Minerva team developed a Grafana-based monitoring framework for continuous tracking of GPU resource utilization across the HPC resource. The monitored workloads include, but are not limited to, GPU-intensive biomedical and AI applications such as DINOv3, Llama 4, PyTorch, TensorFlow, MONAI, NVIDIA Parabricks, GROMACS, DeepLabCut, CryoDRGN, RAPIDS cuDF, and distributed multi-GPU deep learning and genomics workflows. The Grafana dashboard collects GPU utilization metrics from running jobs and visualizes how many GPUs are potentially being wasted by individual users at a given time point. Internally, the monitoring framework uses automated threshold-based analysis, where alerts are generated when a user reserves more than 30 GPUs while maintaining an average GPU utilization below 20% for one hour or longer. When such conditions are detected, automated email notifications containing the user information are sent to the system administration and computational scientist teams for further review. The Minerva computational scientist team then proactively contacts users and provides optimization guidance for their workloads. For example, the monitoring framework identified workloads with extremely low GPU utilization, including one job with an average Streaming Multiprocessor (SM) utilization of only 5% and average GPU memory utilization of 1%. In another case, a user reserved 65 GPUs while maintaining approximately 0% GPU utilization for several days. After repeated unsuccessful attempts to contact the user, the user's GPU allocation was temporarily reduced to zero, immediately returning 65 GPUs to the shared cluster and improving GPU availability for other researchers. Using the Grafana dashboard, support staff can further analyze user-specific utilization trends by selecting individual users from the "Wasted GPUs per user" panel and reviewing historical GPU activity patterns. Based on these analyses, the Minerva team assists researchers in optimizing job scripts, improving data-loading pipelines, and reducing GPU

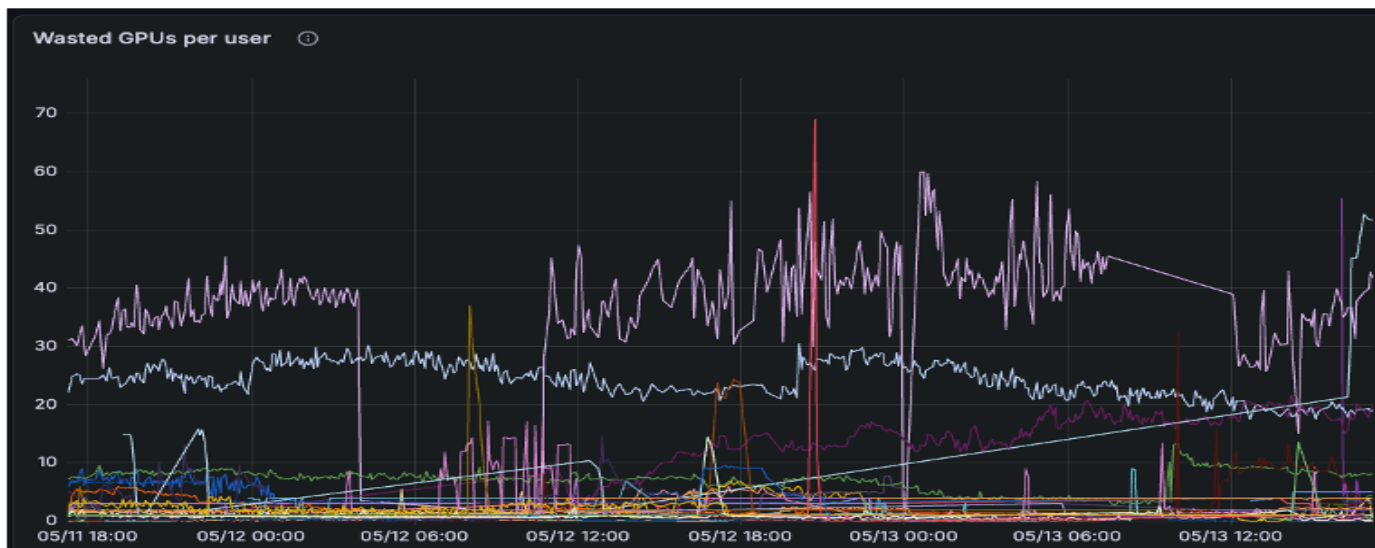


Fig. 3. The Grafana dashboard shows wasted GPUs for different users over time. Each horizontal line represents a user. User names are hidden for security and privacy reasons.

resource waste. This proactive GPU monitoring and user-engagement strategy improves fair-share scheduling efficiency, increases GPU availability for researchers. As shown in Fig. 3, several users were identified with underutilized GPUs over extended periods. The x-axis represents the date and time, while the y-axis represents the number of wasted GPUs associated with individual users over time. Each horizontal line in the figure represents a user. Due to security and privacy concerns, usernames are not shown in Fig. 3. Minerva GPU usage hours were approximately 1,000,000 in 2020, and the number of GPU jobs was about 65,000. Currently, both GPU usage and the number of GPU jobs have grown by approximately 5X compared to 2020.

## V. CONCLUSION AND FUTURE WORK

This paper presented the Minerva high-performance computing system and its role in supporting large-scale biomedical, genomics, imaging, AI, and scientific computing workloads. Minerva combines heterogeneous CPU and GPU resources, including NVIDIA V100, A100, H100, L40S, and B200 accelerators, with high-speed InfiniBand networking, GPFS storage, NVLink interconnects, and a secure HIPAA-compliant environment. HPL Benchmark results demonstrated substantial performance gains on newer GPU architectures, with H100NVL and B200 achieving the highest throughput. Future work will evaluate additional AI and biomedical applications, including large language models, genomics, molecular dynamics, and distributed deep learning, while examining scalability, energy efficiency, storage, and multi-node performance.

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